MONASH University

Department of Econometrics and Business Statistics
ETW3482 DATA MINING AND PREDICTIVE MODELLING

Semester 2, 2026
Assessment 3: Predictive Modelling and Scoring Project
Due Date: Friday, 23 October 2026, 11:55 PM
Weighting: 40% of the final grade

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1. Objective

This assignment assesses your ability to:

1. **Design, implement, and evaluate predictive models using appropriate supervised learning algorithms to support responsible and evidence-based decision-making.**
2. **Communicate insights and recommendations derived from data analysis and predictive modelling clearly and effectively to diverse audiences.**

This assignment extends your Assessment 2 work. You will continue using your cleaned dataset and preprocessing pipeline to construct and explore predictive models, identify a champion model, investigate prediction and model insights, and deploy a predictive solution using SAS Viya.

The emphasis of this assessment is not solely on predictive performance. You are expected to demonstrate an understanding of predictive modelling through systematic model exploration, evidence-based model selection, appropriate model interpretation, and the translation of predictive outputs into actionable organisational strategies.

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2. Overview

In this assignment, you will:

- continue from your Assessment 2 preprocessing pipeline;
- construct **ONE predictive modelling pipeline**;
- explore all predictive models taught in this unit;
- investigate model behaviour through appropriate **hyperparameter exploration**;
- use evidence from model exploration to investigate relationships between the target and input variables;
- select and justify an appropriate performance metric for model comparison;
- identify and justify a **champion model**;
- investigate the prediction performance of the champion model;
- apply appropriate **model interpretability tools** to derive predictor insights;
- deploy the champion model using SAS Model Manager;

- score new observations; and
- translate predictive outputs and insights into actionable strategies.

The report contains two parts:

Part A: Technical Investigation

Audience: Technical Manager

Part B: Business Intelligence and Implementation

Audience: Business Decision Maker

Your writing, evidence and level of technical detail should be appropriate for the intended audience of each part.

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3. Preliminary Requirement (Non-Assessed)

Scoring Dataset Preparation

Create a scoring dataset from your Assessment 2 dataset.

Requirements

- Randomly select **100 observations** from your processed dataset.
- Remove the **target variable**.
- Retain all remaining variables required for scoring.
- Include an **ID variable** if an ID variable does not already exist.

Important

This scoring dataset is not directly assessed. However, incorrect preparation may affect later stages of model deployment and scoring.

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4. Assignment Tasks

Part A: Technical Investigation

Section 1: Dataset and Prediction Objective

Provide a brief overview of:

- dataset context;
- target variable;
- prediction problem; and

- prediction objective.

Purpose

This section provides the necessary project context and is **not directly assessed**.

Avoid unnecessarily repeating material already presented in Assessments 1 and 2.

Section 2: Predictive Model Exploration through Hyperparameter Investigation

Construct a predictive modelling pipeline using all predictive models taught in this unit.

Include:

- Decision Tree
- Random Forest
- Gradient Boosting
- Regression Model
- Neural Network
- Support Vector Machine

For each model, investigate appropriate **hyperparameter settings** and evaluate how changes in these settings affect model behaviour and predictive performance.

The purpose of this investigation is not simply to search for the best-performing model configuration. You are expected to use the hyperparameter exploration to develop an understanding of the relationships between the **target variable and input variables**.

Investigation

Select and investigate relevant hyperparameters for the predictive models.

Your investigation should demonstrate an understanding of:

- what the selected hyperparameters control;
- why the selected hyperparameters and settings are relevant;
- how changing the hyperparameters affects model behaviour;
- how predictive performance changes under different hyperparameter options; and
- what the observed performance differences suggest about the relationships between the target and input variables.

You are **not expected to investigate every available hyperparameter**. Select hyperparameters that are meaningful for understanding model behaviour and the prediction problem.

Discussion

Compare the behaviour and performance of the models under different hyperparameter settings.

Use the combined evidence from the hyperparameter investigations to draw conclusions about the relationships between the target and input variables.

Your conclusions should emerge from the overall exploration process rather than from an isolated model or individual result.

Merely listing hyperparameter settings and corresponding performance values is insufficient. You must explain what the observed differences reveal about the prediction problem.

Higher initiative, systematic investigation and deeper exploration will be rewarded.

Evidence

Include selective screenshots and outputs showing:

- the predictive modelling pipeline;
- relevant hyperparameter configurations;
- model performance under the selected settings; and
- other relevant outputs required to support your investigation.

Section 3: Champion Model Identification

Use the predictive models developed in Section 2 to identify the **champion model**.

Select **ONE appropriate performance metric** to evaluate and compare the candidate models.

Explain

- why the selected performance metric is appropriate for the prediction problem;
- why the metric provides a suitable basis for comparing the candidate models;
- how the candidate models perform according to the selected metric;
- which model is identified as the champion model; and
- why the available evidence supports the champion-model selection.

The champion model should be identified using an appropriate and consistently applied performance metric. Model selection should not be based simply on the model automatically identified by the software.

Important

For this assessment, champion-model identification in this section must be based on the selected **performance metric**.

Performance charts used to investigate prediction behaviour and prediction insights are addressed separately in **Section 4**.

Evidence Required

Include relevant screenshots or outputs supporting the model comparison and champion-model identification.

Part B: Business Intelligence and Implementation

Section 4: Prediction Insights and Model Interpretability

Section 4.1: Prediction Performance Insights

Use the champion model selected in Section 3 to investigate the practical behaviour and usefulness of its predictions.

Select **ONE suitable performance chart** available in SAS Viya, Model Studio that is appropriate to the prediction problem.

You must determine which performance chart is appropriate for your prediction objective.

Explain

- why the selected performance chart is appropriate;
- what the chart reveals about the predictive performance of the champion model;
- what meaningful prediction insights can be derived from the evidence;
- how these insights are relevant to the prediction objective; and
- what the findings imply for the practical use of the predictions.

Do not merely describe the chart, axes or values displayed. Explain the **prediction insights** that can be derived from the evidence.

Section 4.2: Model Interpretability and Predictor Insights

Apply appropriate model interpretability tools to investigate how the champion model behaves across the dataset and to derive meaningful insights about the overall relationships between the input variables and predicted outcomes.

You are **not required to use every interpretability tool**. You must determine which tools are appropriate for the analytical questions being investigated.

Investigation

Use appropriate interpretability tools to identify and investigate the **three most influential predictors**.

Your analysis should determine, where appropriate:

- which predictors have the greatest influence on the champion model's predictions;
- how important numerical predictors are associated with changes in predicted outcomes;
- the direction and pattern of influence observed for important numerical predictors;
- how predicted outcomes differ across the categories of important categorical predictors;
- what the interpretability evidence reveals about the relationships between important predictors and the predicted outcome; and
- what these findings mean in the context of the prediction problem.

The interpretability tool selected should be appropriate for the specific analytical question being investigated.

Evidence and Interpretation

Include relevant outputs from the selected model interpretability tools.

Your discussion should clearly connect:

Interpretability evidence → Target–input insight → Contextual meaning → Potential decision implication

Simply presenting interpretability outputs or identifying important predictors without explaining what the evidence reveals about the predictions is insufficient.

Section 5: Model Deployment and Scoring

Deploy the selected champion model using **SAS Model Manager**.

Tasks

- Register the champion model.
- Import the scoring dataset prepared earlier.
- Apply the deployed model to the scoring dataset.
- Generate predictions for the new observations.
- Produce a scored dataset that retains the observation ID.

Evidence

Include selective screenshots showing:

- champion-model registration;
- the scoring process; and
- the resulting scored output, including the ID variable and relevant prediction output.

Submit

ETW3482_Scored_<StudentID>.csv

The scored dataset should be an output that can be effectively passed to a downstream organisational decision-making or operational process.

Section 6: Actionable Strategy and Implementation Considerations

Assume that the predictive model will be implemented within the organisation represented by your project.

Explain how the **prediction outputs and insights obtained from Section 4** can be translated into practical decisions and actions.

Discuss

- how prediction outputs can be used to support organisational decisions;
- what actions or strategies should follow from relevant prediction outcomes;
- how insights from the influential predictors can inform the proposed strategies;
- how the predictions could be incorporated into an organisational process or workflow;
- who or which organisational function would use the predictions;
- important implementation considerations;

- possible organisational or operational challenges;
- possible consequences of incorrect predictions, where relevant; and
- considerations for responsible and evidence-based use of the predictive solution.

The emphasis of this section is on **operationalising the predictive solution**, rather than describing technical model outputs.

Your proposed strategy should demonstrate a clear connection:

Prediction → Decision → Action → Expected organisational outcome

Strategies should be supported by the prediction and interpretability evidence obtained from the analysis rather than generic recommendations.

Section 7: Conclusion

Provide a concise synthesis of the project.

Summarise:

- key findings from the hyperparameter-based model exploration;
- insights about the relationships between the target and input variables;
- the basis for champion-model selection;
- important prediction-performance insights;
- major findings from the model interpretability analysis;
- strengths and limitations of the predictive solution in addressing the prediction problem; and
- key operational considerations affecting its practical implementation.

The conclusion should synthesise the major findings rather than repeat individual sections of the report.

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5. Report Requirements

Maximum: 3000 words

Excluding

- tables;
- figures;
- screenshots;
- appendix; and
- reflection.

Formatting

- Arial size 12
- Line spacing: 1.5
- Justified alignment
- Page numbering: Bottom centre

Writing Style

Part A:
Technical Manager

Part B:
Business Decision Maker

Tables, figures and screenshots should be selectively used, clearly presented and referred to in the discussion.

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6. Marks Breakdown

Category	Weight (%)
Predictive Model Exploration through Hyperparameter Investigation (Section 2)	25
Champion Model Identification (Section 3)	10
Prediction Performance Insights (Section 4.1)	10
Model Interpretability and Predictor Insights (Section 4.2)	15
Model Deployment and Scoring (Section 5)	10
Actionable Strategy and Implementation Considerations (Section 6)	15
Synthesis and Professional Communication (Section 7 and overall report)	15
Total	**100**

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7. Submission Requirements

Submit the following files via the Moodle assignment submission link before the due date:

Report (PDF):
ETW3482_FinalReport_<StudentID>.pdf

Scored Dataset (CSV):
ETW3482_Scored_<StudentID>.csv

Ensure that the correct files are submitted and that the files can be opened and read successfully.

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8. Important Notes

- The entire predictive modelling, assessment, interpretation, deployment and scoring process must be completed using **SAS Viya Model Studio and Model Manager**.
- Use the data preprocessing pipeline developed in Assessment 2. Modifications to the pipeline are permitted if necessary.
- Use the provided report template for consistency.
- Model interpretability analysis is compulsory regardless of the type of champion model selected.
- Screenshots and outputs should be selective and used as evidence to support the analysis rather than simply document the steps performed.
- Refer to the marking rubric for detailed assessment criteria.

This retains the previous requirement that students continue from the Assessment 2 preprocessing pipeline and use SAS Viya Model Studio and Model Manager for the predictive-modelling and deployment work.

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9. Academic Integrity

All submitted work must comply with the Monash University's academic integrity requirements.

You are responsible for the originality of your work, the accuracy of your analysis and interpretation, and the appropriate acknowledgement of sources used in preparing the submission.

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10. Use of Generative Artificial Intelligence

Any use of generative artificial intelligence tools must comply with the Monash University's policies and the unit-specific instructions provided on Moodle.

You remain responsible for:

- the originality of your work
- the accuracy of all statements
- the credibility of sources and references
- the consistency of the submitted work
- your ability to explain and defend the submitted work

Submitting AI-generated content without appropriate critical evaluation, acknowledgement where required, or personal understanding may constitute inappropriate academic practice.

The teaching team may ask a student to explain or clarify aspects of the submitted work where authorship or understanding needs to be verified.

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11. Late Submission, Extensions and Special Consideration

Unless an extension or special consideration has been granted, a late-submission penalty of **5% of the available marks will be applied for each day (24-hour period), or part thereof, that the assignment is overdue.**

Assignments submitted more than **seven calendar days after the due date will not be marked**.

Students requiring an extension or special consideration must apply through the appropriate Monash University system. Informal communication to the teaching team does not constitute an approved extension.

All late submissions, extensions and special-consideration applications will be handled by SEBS in MUA in accordance with Monash University policies and procedures.

_____The End_____